

QSKILL INTERNSHIP HOUSE PRICE PREDICTION

Machine Learning Project

Final Project Report

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1. Objective

The objective of this project is to develop a Machine Learning model that predicts house prices using housing-related features. The project uses the California Housing dataset and applies Linear Regression for prediction.

2. Introduction

House price prediction is a regression problem in which a model learns the relationship between housing features and a target house value. In this project, the dataset is explored, divided into training and testing data, and used to train a Linear Regression model. The trained model is then evaluated using R² Score, Mean Absolute Error (MAE), and Mean Squared Error (MSE).

3. Dataset

The notebook uses the California Housing dataset loaded through scikit-learn. The dataset contains 20,640 records and 9 columns. Eight columns are input features and MedHouseVal is the target variable.

Type	Details	Count
Dataset	California Housing	20,640 records
Input features	MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude	8
Target	MedHouseVal	1

4. Dataset Inspection

The notebook reports 20,640 rows and 9 columns. The listed columns are MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude, and MedHouseVal. The missing-value check shows 0 missing values for every column.

5. Data Exploration

Descriptive statistics were generated using df.describe(). The notebook also includes a house-price distribution histogram and a correlation heatmap to understand the distribution of the target and relationships among numerical variables.

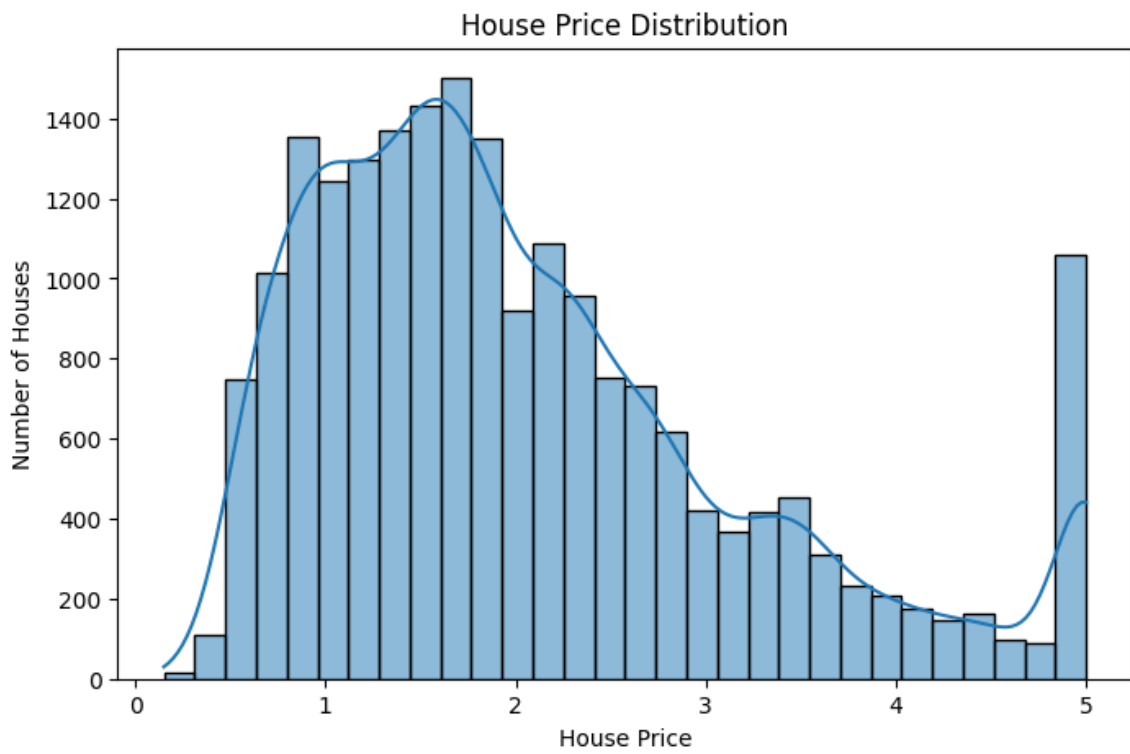


Figure 1: House Price Distribution

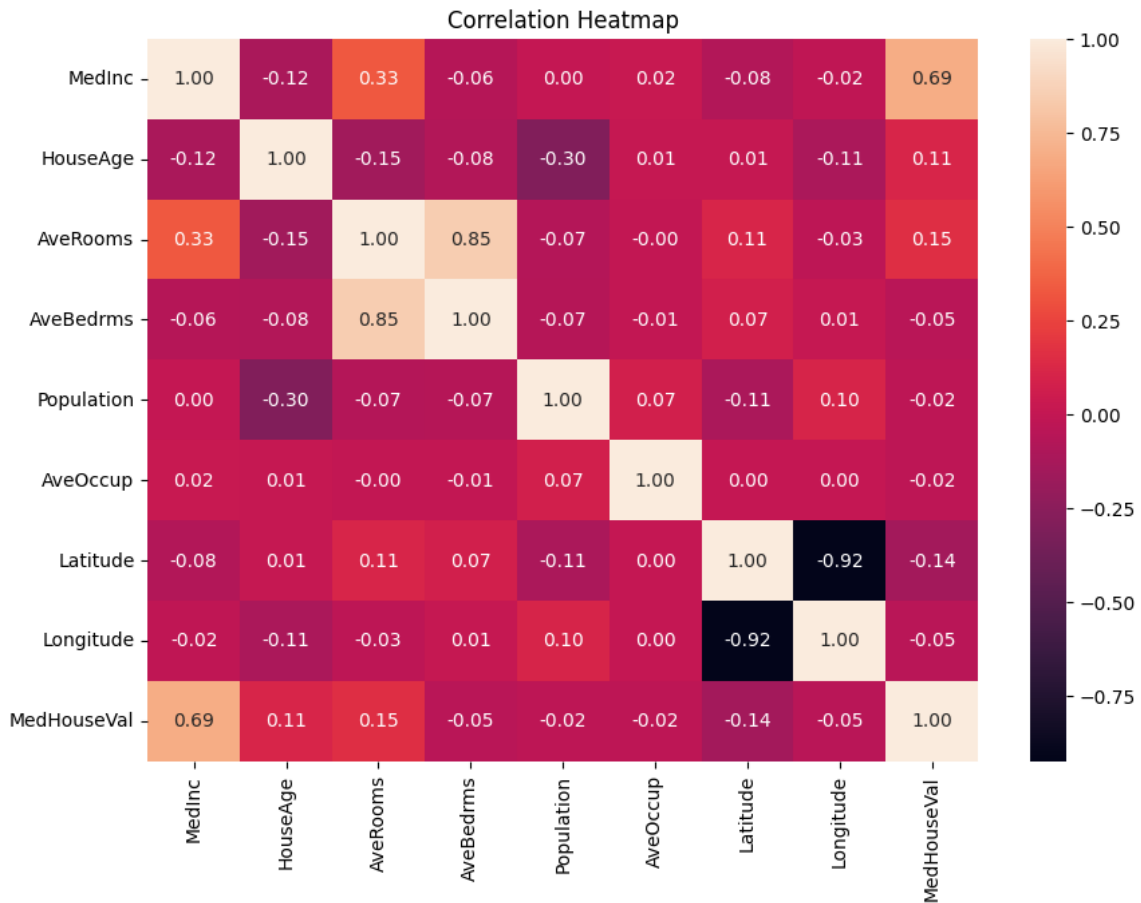


Figure 2: Correlation Heatmap

6. Input and Target Variables

The target column MedHouseVal is removed from the input data to create X, while MedHouseVal is assigned to y. Therefore, the model learns to predict MedHouseVal from the remaining eight housing features.

7. Train-Test Split

The data was divided using train_test_split with test_size=0.2 and random_state=42. The resulting dataset contains 16,512 training samples and 4,128 testing samples.

8. Machine Learning Model

Linear Regression was selected as the prediction algorithm. The model was trained using the training features and target values.

9. Model Prediction

After training, the model generated predictions for the test dataset. The first ten predicted values recorded in the notebook are: 0.7191, 1.7640, 2.7097, 2.8389, 2.6047, 2.0118, 2.6455, 2.1688, 2.7407, and 3.9156.

10. Model Evaluation

Metric	Result
R ² Score	0.5758 (57.58%)
Mean Absolute Error (MAE)	0.5332
Mean Squared Error (MSE)	0.5559

The R² Score of 0.5758 indicates that the Linear Regression model explains about 57.58% of the variation in the test-set target values. MAE and MSE provide additional measures of prediction error.

11. Actual vs Predicted Prices

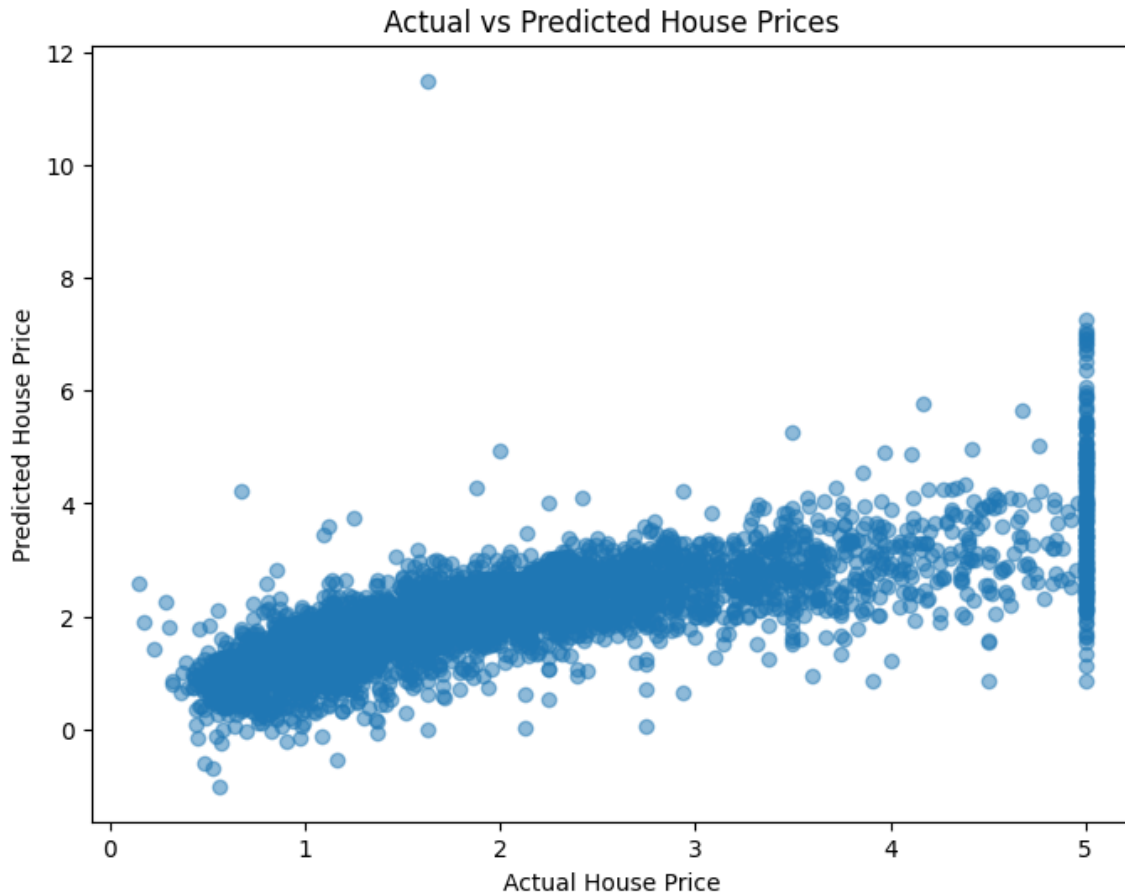


Figure 3: Actual vs Predicted House Prices

12. New House Prediction

A new house record was supplied to the trained model using the eight input features. The notebook produced a predicted value of 2.5227.

The notebook also generated a comparison table containing actual and predicted prices for the first ten test samples.

13. Results

- California Housing dataset successfully loaded and explored.
- No missing values were found in the dataset.
- The data was split into 16,512 training samples and 4,128 testing samples.
- Linear Regression was successfully trained.
- The model achieved an R^2 Score of 0.5758 (57.58%).
- The model recorded MAE = 0.5332 and MSE = 0.5559.
- A new house prediction produced the value 2.5227.

14. Conclusion

The House Price Prediction project successfully demonstrates a complete Machine Learning regression workflow. The California Housing dataset was used for training and testing, and Linear Regression was applied to predict house values from housing features. The model was evaluated using R^2 Score, MAE, and MSE, and it was also used to generate a prediction for a new house record.

15. Tools and Technologies

Python, Google Colab, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn were used in the notebook.

16. Project File

House_Price_Prediction.ipynb